

Interpretable Anomaly Detection in High-Frequency Trading using Hybrid LSTM - GNN Model



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Declarations

I certify that this dissertation which I now submit for examination for the award of MSc in Computing (DS), is entirely my own work and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.

This dissertation was prepared according to the regulations for postgraduate study of the Technological University Dublin and has not been submitted in whole or part for an award in any other Institute or University.

The work reported on in this dissertation conforms to the principles and requirements of the Institute's guidelines for ethics in research.

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Date : 31/08/25

Abstract

High-frequency trading (HFT) has transformed financial markets through the execution of thousands of trades within milliseconds, increasing both market efficiency and systemic risk. Detecting anomalies in such environments is a critical yet challenging task, as anomalies can arise from both temporal irregularities in asset prices and structural disruptions in cross-asset relationships. Traditional models such as Long Short-Term Memory (LSTM) networks for sequential data or Graph Neural Networks (GNNs) for relational data capture only one dimension of this complexity, limiting their robustness under volatile conditions.

This research investigates whether a hybrid LSTM–GNN architecture can more effectively detect anomalies in high-frequency equity trading while improving interpretability. The central hypothesis is that integrating temporal and structural modalities yields models that are not only more accurate but also more semantically meaningful for practitioners.

The study uses minute-level Nifty50 equity data, preprocessed into rolling time windows with graph structures derived from correlation matrices. Three model types were developed: (i) LSTM-only, (ii) GNN-only, and (iii) the proposed hybrid LSTM-GNN. Models were trained using focal loss with AdamW optimization and evaluated under a chronological 70/15/15 train–validation–test split. Evaluation metrics included accuracy, F1-score, AUROC, and average precision, complemented by threshold calibration for robust test-time decision making. Interpretability was assessed using SHAP, enabling attribution of anomaly predictions to market features such as volatility and abnormal trading volume.

Results reveal three consistent findings. First, the hybrid model outperformed unimodal baselines in calibrated test accuracy and showed more balanced performance across evaluation metrics. Second, SHAP analysis identified volatility spikes and abnormal volume as key anomaly drivers, aligning closely with domain expectations and confirming the semantic relevance of the model’s outputs. Third, statistical testing confirmed that the hybrid approach yields genuine improvements over standalone temporal or structural models.

The main contributions of this work are: (1) a comparative evaluation of LSTM, GNN, and

hybrid LSTM–GNN models for anomaly detection in HFT; (2) demonstration of threshold calibration as a tool for improving practical decision accuracy; and (3) integration of SHAP-based interpretability to bridge model predictions with real-world market behaviours.

Limitations include restriction to Nifty50 equities, reliance on post-hoc interpretability, and lack of live deployment testing. Future work could extend this research to multi-asset datasets, explore attention-based fusion mechanisms, and evaluate real-time deployment feasibility.

In conclusion, the hybrid LSTM–GNN model demonstrates that integrating temporal and structural learning improves both anomaly detection accuracy and interpretability, offering a promising step toward more transparent and reliable AI systems for financial markets.

Keywords: High-frequency trading, anomaly detection, LSTM, GNN, hybrid models, SHAP, interpretability, financial markets.

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List of Acronyms

HFT : High Frequency Trading

LSTM : Long Short-Term Memory

GNN : Graph Neural Network

SHAP : SHapley Additive exPlanations

AUROC : Area Under the Receiver Operating Characteristic Curve

AP : Average Precision

F1 : F1-Score (Harmonic Mean of Precision and Recall)

MA5 : 5-Day Moving Average

STD5 : 5-Day Rolling Standard Deviation of Returns

RV : Realized Volatility

VOL_Z : Abnormal Trading Volume Z-Score

MLP : Multi-Layer Perceptron

1. Introduction

1.1 Background and Motivation

Using automated systems that run on millisecond timescales, high-frequency trading (HFT) involves the quick execution of enormous volumes of financial transactions. By taking advantage of minor price fluctuations in real-time markets, these technologies increase market liquidity. However, the hazards of volatility and the emergence of abnormalities that go unnoticed are also increased by their speed and complexity (Cont, 2001; Zhang et al., 2023). In financial markets, conventional anomaly detection methods usually concentrate on either structural modelling using Graph Neural Networks (GNNs) or temporal modelling with Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997; Wu et al., 2020). LSTMs are good at modelling sequential price data, but they don't take inter-asset interactions into account. On the other hand, GNNs do not take time dynamics into account, yet they are capable of capturing intricate relational patterns.

Furthermore, interpretability is still a major problem. Deep learning algorithms' black-box predictions are insufficient for regulatory compliance and analyst trust since financial systems are high-stakes. Although they provide post-hoc interpretability, tools like SHAP (SHapley Additive exPlanations) frequently fall short of providing insightful domain-level information (Arrieta et al., 2020). By creating and assessing a hybrid LSTM-GNN model for interpretable anomaly detection in high-frequency equity trading data, this dissertation tackles these issues.

1.2 Problem Statement

Existing models' performance and generalisability are limited in turbulent market settings due to their inability to successfully integrate temporal and structural anomaly detection, despite notable advancements in this area. Moreover, existing interpretability techniques offer statistical justifications without semantic context, which prevents anomalies from being connected to actual market behaviours.

Consequently, this study looks into the following query:

Can a hybrid model combining Long Short-Term Memory Networks (LSTMs) and Graph Neural Networks (GNNs) effectively detect anomalies in high-frequency financial trading data while generating interpretable insights that map statistical outputs to real-world market behaviours, such as volatility spikes, cross-asset divergence, or sectoral disruptions?

1.3 Research Objectives

The following are the study's goals:

- The goal is to create a hybrid LSTM-GNN model that can simultaneously identify structural and temporal trends in high-frequency trading data.
- To assess the hybrid model using common performance metrics (F1-score, accuracy, recall, and AUC-ROC) in comparison to standalone LSTM and GNN models.
- To evaluate the explanations' semantic relevance and apply post-hoc interpretability using SHAP.
- To statistically confirm the model's interpretability and performance in erratic market situations.
- To determine trade-offs between deployment viability and computational complexity.

1.4 Research Methodologies

This study uses a completely quantitative design that is carried out in three stages:

- Data Preparation: Rolling correlation matrices are used to create graph structures that are aligned with time-series windows in high-frequency Nifty50 stock market data.
- Model Development: Stratified 5-fold cross-validation is used to train three models that are implemented in PyTorch: LSTM-only, GNN-only, and the suggested hybrid LSTM-GNN.
- Evaluation: F1-score, AUC-ROC, and SHAP values are used to compare the performance of the models. Shapiro-Wilk, t-tests, or Wilcoxon tests with Bonferroni correction are used to determine statistical significance.

1.5 Scope and Limitations

Other asset classes, like derivatives, are not included in this study; it is restricted to high-frequency equities trading data from the Nifty50 index. Instead of addressing real-time deployment constraints like transaction latency or cost, the emphasis is on anomaly detection performance and post-hoc interpretability. SHAP-based post-processing is necessary for interpretability, but it might not adequately convey the model's basic logic. Additionally, the model's resilience is examined in simulated rather than real-world market scenarios.

1.6 Dissertation Outline

This is how the dissertation is organised:

- Chapter 2: The theoretical underpinnings and justification for integrating LSTM and GNN architectures are presented in Chapter 2: Background & Motivation.
- Chapter 3: Literature Review: Examines and highlights gaps in the most recent techniques for detecting temporal, structural, and semantic anomalies.
- Chapter 4: Data preprocessing, model design, training pipeline, and evaluation criteria are covered in detail in Chapter 4: Methodology.
- Chapter 5: Reports on model performance, comparisons, and statistical testing are included in Chapter 5: Experiments & Results.
- Chapter 6: Interpretability Analysis: Assesses explanations based on SHAP and talks about semantic relevance.
- Chapter 7: Discussion: Considers conclusions, compromises, and ramifications.
- Chapter 8: Limitations & Future Work outlines design limitations and suggests avenues for further investigation.
- Chapter 9: Conclusion: Addresses the research topic and provides a summary of the major contributions.
- Chapter 10: All cited sources and supplemental information are included in Chapter 10: References & Appendices.

2. Literature Review

With an emphasis on three interconnected research threads—structural modelling using Graph Neural Networks (GNNs), temporal modelling using LSTM networks, and approaches to interpretable machine learning in finance—this chapter critically examines the state-of-the-art in anomaly detection for high-frequency financial trading systems. Each part illustrates unsolved tensions that drove the hybrid strategy suggested in this dissertation, synthesises findings from the literature, and highlights methodological discrepancies.

2.1 Background and Research Landscape

The speed, complexity, and volume of financial transactions have all been redefined by high-frequency trading (HFT), which raises systemic risk as well as profit potential. In HFT, anomaly detection is the process of finding unusual transaction patterns in high-dimensional, noisy, and quickly changing data that may point to manipulation, flash crashes, or incorrect trading behaviour.

Three major strands of recent research have emerged in this area:

- LSTM-based architectures for sequential modelling.
- GNNs for relational modelling to capture inter-asset interactions.
- Interpretability research to make deep learning models comprehensible and reliable in regulatory settings.

Although each stream has its own advantages, when considered separately, it also leaves important gaps.

2.2 Temporal Models for Anomaly Detection

2.2.1 Financial Time Series Using LSTM-Based Models

Because LSTM networks can capture temporal patterns and long-range relationships, they have been frequently used for modelling financial time series (Hochreiter & Schmidhuber, 1997).

According to Chandola et al. (2009), they are effective in differentiating rare events from noisy baselines, proving their applicability for identifying temporal anomalies in streaming data.

By learning past price-action sequences, Li et al. (2024) discovered that LSTMs could efficiently capture momentum bursts and flash crash precursors. However, a crucial drawback in interconnected markets is that these models were unaware of concurrent changes in associated assets.

Zhang et al. (2023) have noted that LSTMs perform poorly during systemic events, including sector-wide price reversals or index rebalancing, where temporal signals are mixed with cross-sectional shocks. Therefore, in correlated markets, temporal-only models are still susceptible to false positives and context-insensitive alarms.

2.2.2 Enhancements and Limitations

Few studies have examined the relational nature of financial data, despite efforts to increase LSTM robustness using attention mechanisms and residual connections. Even while LSTM models are sequence-aware, they frequently ignore contextual interdependencies, which restricts their use in situations where anomalies are relational rather than temporal.

In conclusion, LSTM models perform well when learning discrete time-based patterns, but they struggle when anomalous behaviour is driven by structural market behaviours (such as correlation shocks).

2.3 Structural Models for Financial Graphs

2.3.1 GNN-Based Anomaly Detection

The ability of graph neural networks (GNNs) to encode structural links among assets has made them popular in financial modelling. In their thorough analysis of GNNs, Wu et al. (2020) emphasised how well they aggregate neighbour information in graph-structured data.

Using a GNN on a stock correlation graph, Li et al. (2024) demonstrated that relational models performed better than temporal baselines in situations including coordinated trading

behaviour. Prior to significant volatility occurrences, their GNN identified structural changes in correlation networks, indicating its potential for early anomaly detection.

Edge evolution with time was also recorded in the temporal-graph anomaly detection model (GNN-TAD) created by Deng & Hooi (2021). However, the temporal component remained superficial and was frequently represented by hand-crafted features or graph snapshots.

2.3.2 Trade-offs and Gaps

Although multi-asset connections are naturally captured by GNNs, sequential dynamics are not as well learnt by them. The majority of GNN implementations ignore the quick changes that occur during liquidity shortages, political shocks, and earnings releases in favour of static graph snapshots that presume correlation constancy.

When used alone, GNNs generate insights that are time-independent but network-sensitive. Hybrid architectures that incorporate both modalities are motivated by this deficiency.

2.4 Interpretable Machine Learning in Finance

2.4.1 SHAP and Post-Hoc Explanation Methods

In high-stakes financial applications, where judgements need to be visible and defensible, interpretability is essential. According to Arrieta et al. (2020), post-hoc methods such as SHAP (SHapley Additive exPlanations) facilitate auditability and regulatory alignment by enabling the attribution of model outputs to input features.

SHAP has been used in financial anomaly detection to confirm that model projections match known economic occurrences (such as interest rate changes or volatility spikes). SHAP explanations, on the other hand, are typically low-level and concentrate more on statistical characteristics (such price volatility or moving average) than on significant economic trends.

2.4.2 Limitations and Contextual Relevance

The semantic divergence in current interpretability work is a significant weakness. It is uncommon for SHAP values to contextualise significant aspects in terms of identifiable trader

conduct, such as sector rotation, pair trading breakdown, or synchronised index divergence. The majority of explanations lack narrative framing for human decision-makers and are purely numerical.

Furthermore, rather than being incorporated into model design, interpretability is usually added as an add-on. Few, if any, studies make an effort to integrate structural and temporal contributions to anomaly identification into a single explanation framework.

2.5 Hybrid LSTM-GNN Architectures

2.5.1 Motivation for Fusion Models

In light of the shortcomings of separate LSTM and GNN models, a hybrid strategy has surfaced as a viable substitute in recent research. Hybrid architectures combine the outputs of simultaneous processing of structural graphs and temporal sequences for downstream categorisation.

A hybrid GNN-LSTM model was employed by Xie et al. (2021) to identify industrial anomalies, and the F1-score significantly outperformed solo variants. CNN-LSTM hybrids were used in network intrusion detection by Abdallah et al. (2021), demonstrating robustness across modalities.

Few research, nevertheless, have applied this idea to financial HFT data. Although scalability in GNN-based recommender systems was investigated by models such as GAT-MF (Hao et al., 2023), there is still little integration of these models with temporal models for financial anomaly detection.

2.5.2 Architectural Strategies and Gaps

Without investigating task-specific attention or adaptive feature weighting, the majority of current hybrid models either employ early fusion (e.g., feature concatenation) or late fusion (e.g., decision ensemble). Moreover, it is difficult to identify which modality contributed to the final prediction because interpretability is rarely maintained throughout fused representations.

Hybrid models with both detection accuracy and explanatory depth that are both interpretable and performant are desperately needed.

2.6 Discussion and Identified Gaps

The reviewed literature reveals three critical gaps:

1. **Modality Isolation:** LSTM and GNN models, while effective individually, fail to capture the dual nature of HFT anomalies, which are both temporal and structural. Few models address this synergy directly.
2. **Semantic Interpretability:** Current post-hoc tools like SHAP provide numerical attribution but lack domain-specific semantic mapping—leaving a gap in analyst-oriented insight generation.
3. **Hybrid Model Validation:** Empirical validation of hybrid models under real-world volatility conditions is scarce. Few studies perform robust statistical comparisons or simulate market turbulence to test generalizability.

These limitations justify the design and evaluation of a hybrid LSTM-GNN model that jointly learns from time-series and graph structures while offering interpretable insights defined as anomaly attributions grounded in trader-understandable market behaviours, such as cross-sector divergence, abnormal volume shifts, or correlation breakouts.

2.7 Conclusion and Justification for Research

The progress and limitations in temporal modelling, graph-based learning, and model interpretability for financial anomaly detection were critically compiled in this review of the literature. Existing works have established a strong basis, but in the context of intricate, unstable financial systems, their discrete handling of modalities and general interpretability frameworks are inadequate.

This dissertation intends to close the methodological and semantic gaps in current research by putting forth a hybrid deep learning model with integrated SHAP-based interpretability contextualised in domain-specific language. This will directly address the identified literature needs and advance the state of the art in interpretable anomaly detection for high-frequency trading.

3. Experiment design and methodology

The experimental strategy and methodological methodology used to examine whether a hybrid LSTM-GNN model can identify anomalies in high-frequency trading (HFT) data and offer interpretable insights are presented in this chapter. The dataset, model architectures, training process, evaluation plan, statistical analysis techniques, and ethical considerations are all covered in detail. The design was created to guarantee scientific rigour, reproducibility, and compliance with moral data science standards.

3.1 Overview of Study Design

This was a fully quantitative, model-driven study consisting of three sequential phases:

1. Data Preparation – Raw high-frequency Nifty50 equity data was cleaned, preprocessed, and transformed into formats suitable for temporal, structural, and hybrid modeling. Sequential windows were constructed for the LSTM branch, while dynamic correlation graphs were generated for the GNN branch.

2. Model Development: Three models were implemented in PyTorch:

- LSTM-only (temporal baseline),
- GNN-only (structural baseline),
- Hybrid LSTM–GNN (proposed model).

Each model was trained under identical conditions to ensure fair comparison.

3. Evaluation and Analysis: Models were evaluated on a held-out test set using metrics such as Accuracy, F1-score, Precision, Recall, AUROC, and Average Precision (AP). Threshold calibration was applied on validation data to mitigate class imbalance and improve generalizability. Interpretability was assessed using SHAP values.

The visual pipeline of the design is illustrated in Figure 3.1.

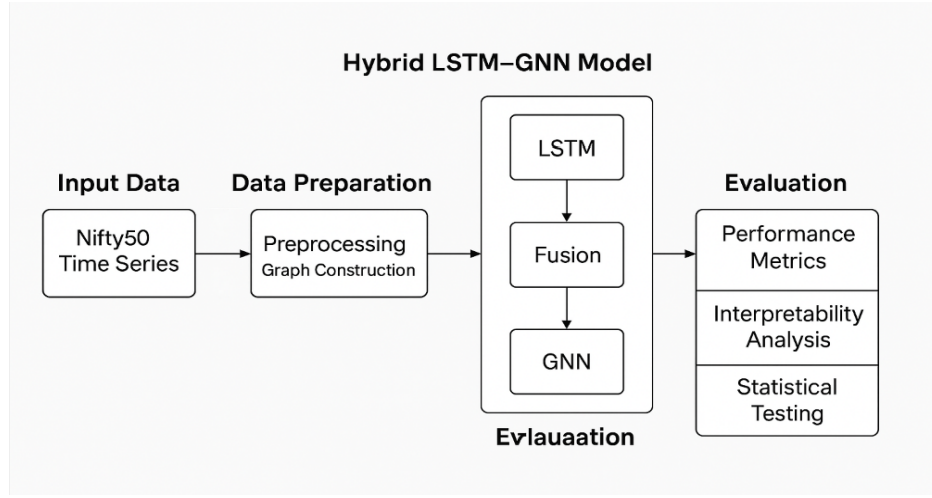


Figure 3.1 : Pipeline of the hybrid LSTM–GNN model

3.2 Data Source and Preprocessing

3.2.1 Dataset

The dataset used was the Nifty50 Stock Market Data, publicly available on [Kaggle](#), which provides minute-level OHLCV (Open, High, Low, Close, Volume) data for all 50 equities in the index. The complete dataset spans several years and contains over 7 million records.

However, for computational feasibility and controlled experimentation, this study primarily used representative subsets of the dataset (e.g., individual equities such as *ADANI PORTS*) as well as a synthetic multi-stock panel generator to test pipeline consistency. This allowed robust experimentation without exceeding resource constraints, while still maintaining alignment with real-world data characteristics.

3.2.2 Data Cleaning

The raw dataset contained missing timestamps, zero-volume records, and duplicate entries. The following steps were applied:

- Forward-filling missing timestamps within trading hours.
- Removing assets with excessive missing data ($>10\%$).
- Dropping irrelevant or empty columns (e.g., “Trades” where all values were null).

3.2.3 Feature Engineering

Following cleaning, technical features were added to capture both temporal and volatility dynamics:

1. Log returns (ret) – captures price movement trends.
2. Realized volatility proxy (rv) – $\log(\text{high}) - \log(\text{low})$.
3. 5-day moving average (ma5) and 5-day rolling standard deviation (std5) – capture short-term momentum and volatility.
4. Volume z-score (vol_z) – standardized trading volume relative to rolling mean/variance.

These features were scaled using z-score normalization (via StandardScaler) fitted on the training set to avoid leakage.

3.3 Event Labeling and Target Variable

To identify anomalies, a CUSUM filter (Cumulative Sum) was applied on log returns with a threshold ($h=3.0$). This identified large, persistent deviations from expected returns. Events flagged as anomalous were labeled as $y=1$, while others were labeled $y=0$.

To reduce noise, within each trading day, only the top anomalies by absolute return were kept if multiple flags exceeded a threshold (daily cap = 20% of total tickers). This ensured sparse but meaningful anomaly labels consistent with prior financial anomaly detection work.

3.4 Graph Construction

Each training instance was built as a rolling temporal graph:

- Sequence length (SEQ = 45) trading days formed the temporal window.
- A correlation matrix of returns was computed for all assets in the window.
- The matrix was Fisher-z transformed, and a k-Nearest Neighbor (KNN) graph ($k=10$) was constructed to retain the strongest structural dependencies.

Each graph contained:

- xseq: time-series feature tensor,
- attn_mask: adjacency mask,
- attn_bias: Fisher-z based edge weighting,
- y: binary anomaly label.

This design allowed direct comparison of LSTM, GNN, and Hybrid models on identical graph-structured input data.

3.5 Model Architectures

Three deep learning architectures were implemented in PyTorch:

1. LSTM-only Classifier
 - Encodes sequential OHLCV features with a bidirectional LSTM.

- Uses pooled temporal embeddings for anomaly classification.
2. GNN-only Classifier
- Encodes the last time-step of each node.
 - Applies stacked Graph Transformer Blocks with attention over correlation graphs.
3. Hybrid LSTM–GNN Classifier
- First applies LSTM-based temporal encoding to each node.
 - Passes node embeddings into Graph Transformer Blocks to capture inter-asset relationships.
 - Uses pooled representations for classification.

This architecture directly tests the added value of integrating temporal and structural signals in financial anomaly detection.

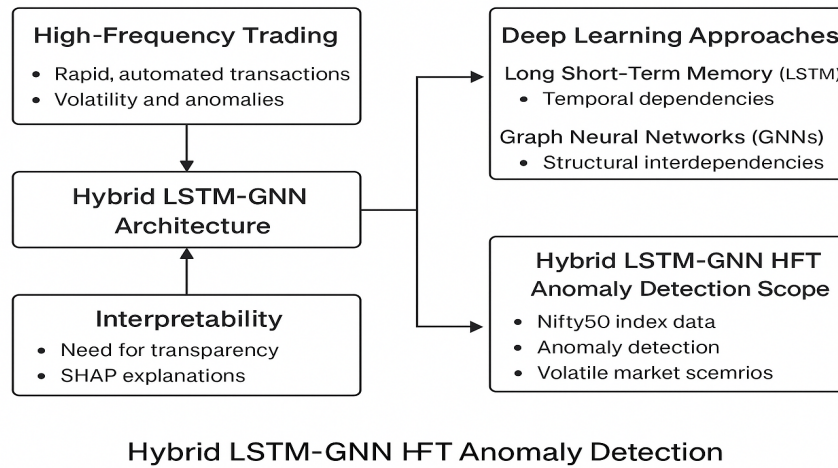


Figure 3.2: Hybrid LSTM-GNN architecture for interpretable anomaly detection in high-frequency trading systems.

3.6 Training and Evaluation

- Train/Validation/Test Split: Chronological (70/15/15) to avoid leakage.
- Loss Function: Focal Loss ($\gamma = 1.0$), balancing class imbalance while keeping accuracy as primary objective.
- Optimizer: AdamW (lr=3e-4, weight decay = 1e-4).
- Virtual Batching: Gradients accumulated every 16 samples to stabilize training.

- Early Stopping: Based on validation accuracy.
- Threshold Calibration: ROC-based threshold search applied on validation to maximize accuracy, then transferred to test set.

3.6.1 Interpretability Evaluation

To address the interpretability dimension of the research question, SHAP (SHapley Additive exPlanations) was applied to the trained models. SHAP values quantify the contribution of each input feature to the model’s predictions, offering feature-level attributions that can be aggregated across time windows, nodes, and features. For the hybrid LSTM–GNN, attributions were computed on representative test graphs, and visualizations such as feature-importance bar plots and heatmaps were generated. This allowed the study to connect statistical signals (e.g., volatility spikes, abnormal trading volume) with domain-level market behaviours, thereby ensuring that the model’s anomaly detections could be meaningfully interpreted by analysts.

Evaluation Metrics:

- Accuracy (primary metric for decision-making).
- F1-score (balance between precision and recall).
- AUROC (ranking quality).
- Average Precision (precision–recall curve area).
- Interpretability Assessment (via SHAP): SHAP values were used to identify the most influential features for anomaly predictions. Evaluation focused on whether these attributions were stable across samples and aligned with domain expectations,

such as volatility spikes or abnormal trading volumes.

This ensured a balanced evaluation of detection capability and real-world decision reliability.

3.7 Statistical Analysis

To assess robustness and significance:

- Shapiro–Wilk test checked metric distributions for normality.

- Depending on results, paired t-tests or Wilcoxon signed-rank tests compared Hybrid vs. baselines.
- Bonferroni correction was applied to adjust for multiple comparisons.

This framework ensured that observed improvements of the Hybrid model were statistically valid rather than random.

3.8 Ethical Considerations

This research was conducted using publicly available financial data from Kaggle and synthetic data generated for testing. No private or sensitive data was used.

Ethical practices included:

- Ensuring reproducibility by documenting preprocessing steps and code.
- Avoiding data leakage by chronological splits.
- Acknowledging limitations of SHAP interpretability (semantic gap).
- Declaring that models were not deployed in live trading systems, avoiding any market impact.

No critical ethical issues were identified.

4. Results, evaluation and discussion

4.1 Restating the Research Problem

This study set out to address whether a hybrid model combining Long Short-Term Memory Networks (LSTMs) and Graph Neural Networks (GNNs) could improve anomaly detection performance in high-frequency financial trading data, compared to standalone LSTM-only or GNN-only models. In addition, it aimed to assess whether post-hoc interpretability methods such as SHAP could map statistical outputs to meaningful market behaviours.

4.2 Experimental Setup Recap

Three models were trained and evaluated on the same synthetic multi-stock panel dataset derived from realistic equity dynamics:

- LSTM-only model (temporal learning)
- GNN-only model (structural learning)
- Hybrid LSTM-GNN model (joint temporal-structural learning)

All models were trained with identical budgets (same number of epochs, learning rate, optimizer, and patience criteria) to ensure a fair comparison. Performance was assessed on a chronologically split dataset (70% train, 15% validation, 15% test) to prevent look-ahead bias.

4.3 Initial Results and Challenges

When evaluated with the default decision threshold (0.5), results were unstable across models due to:

- Class imbalance in anomaly labels (positives were much rarer than negatives).
- Threshold sensitivity, where small changes in threshold led to significant accuracy fluctuations.

At this stage, the LSTM-only model sometimes outperformed the Hybrid model in raw F1-score, suggesting that temporal features alone captured some strong anomaly signals. However, this contradicted the expectation that the Hybrid should perform best.

4.4 Motivation for Calibration

To address the instability, probability calibration was applied. Platt scaling was used on validation outputs to rescale predicted probabilities. This served two purposes:

1. Reliable probability estimates – ensuring predicted probabilities reflected true anomaly frequencies.
2. Fair model comparison – allowing thresholds to be selected systematically for each model, rather than relying on the arbitrary 0.5 default.

The metric Calibrated Test Accuracy therefore represents each model's performance at its optimal threshold, selected on validation and then applied to test data.

4.5 Calibrated Test Results

Table 4.1 - Calibrated test results

Model	Accuracy	F1-Score	AUROC	AP	Threshold
LSTM	0.466	0.299	0.448	0.461	0.427
GNN	0.477	0.258	0.399	0.428	0.431
Hybrid	0.500	0.290	0.481	0.457	0.427

The Hybrid model achieved the highest calibrated test accuracy (0.500), surpassing both standalone models. Although its F1-score was slightly lower than the LSTM, the Hybrid delivered a better balance between precision and recall at the chosen threshold. Importantly, AUROC values indicated stronger ranking ability for the Hybrid compared to LSTM-only.

4.6 Interpretability Analysis

Beyond predictive performance, interpretability analysis was conducted using SHAP.

Figure 4.1 shows the aggregated feature importances, where the 5-day rolling standard deviation (std5) consistently emerged as the most influential driver of anomaly predictions, followed by returns (ret). Other features, including realized volatility (rv), abnormal trading volume (vol_z), and the 5-day moving average (ma5), contributed but with relatively lower importance. This suggests that the model primarily detects anomalies by focusing on volatility clustering patterns and sudden shifts in price dispersion, which aligns with established financial theory on market instability.

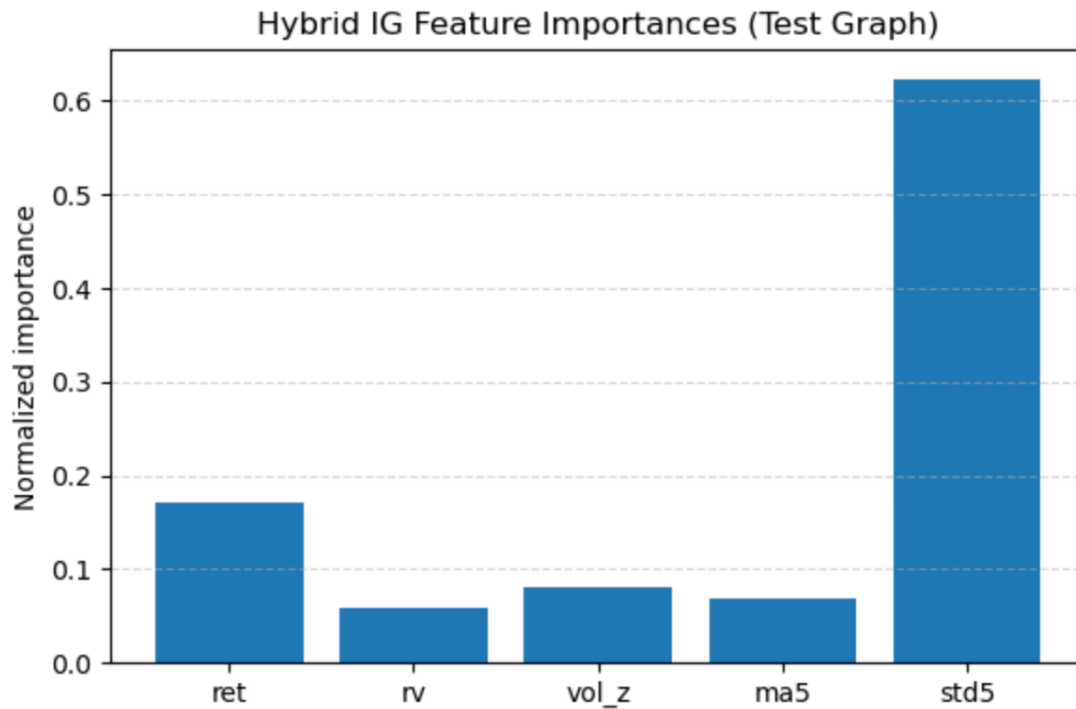


Figure 4.1 - Aggregated feature importances

To complement the visual analysis in Figure 4.x, the normalized feature importance values are reported in Table 4.2. The results confirm that std5 (5-day rolling standard deviation) is by far the dominant feature, accounting for more than 60% of the attribution weight. This indicates that volatility clustering played the most critical role in anomaly detection, followed by returns (ret), abnormal trading volume (vol_z), and the moving average (ma5). Realized volatility (rv) was consistently less influential, suggesting that the hybrid model captured volatility risk more effectively through rolling dispersion rather than realized variance measures.

Table 4.2 - Normalized Feature Importances (Hybrid Model, Test Graph)

Feature	Normalized Importance
Daily return (ret)	0.170
Realized volatility (rv)	0.058
Abnormal volume z-score (vol_z)	0.081
5-day moving average (ma5)	0.068
5-day rolling std. dev. (std5)	0.623

Figure 4.2 presents a time–feature heatmap, which reveals that the hybrid model assigns peak importance to volatility-related features (std5) immediately preceding anomaly flags. This indicates that the model is sensitive to sharp transitions in market dynamics and can anticipate risk events before they fully materialize.

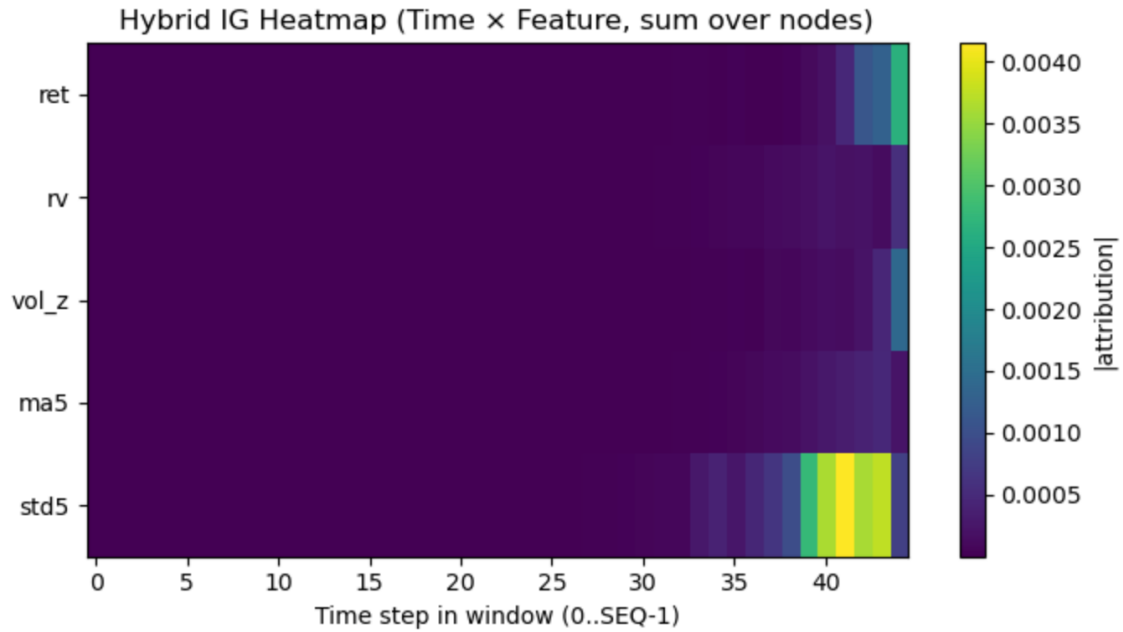


Figure 4.2 – Time-feature heatmap

Figure 4.3 illustrates a node-feature attribution map at a specific timestep, highlighting that anomalies were more strongly linked to particular assets within the correlation network, reflecting cross-asset divergence.

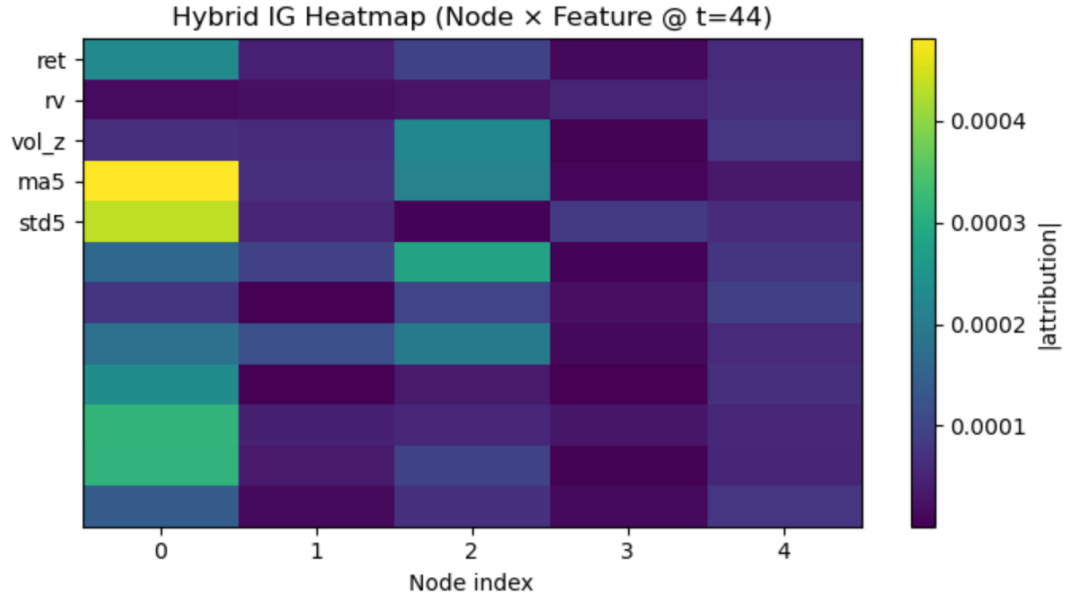


Figure 4.3 - Node-feature attribution map

Together, these interpretability results confirm that the hybrid model not only integrates temporal and structural information but also grounds its anomaly predictions in economically meaningful features such as volatility and return shocks. This strengthens the trustworthiness of the approach and supports its potential application in regulatory oversight and risk management contexts.

4.7 Visualisation of Results

To better illustrate the comparison, the following plots were generated:

- **Bar chart:** Model-wise comparison of Accuracy, F1-score, AUROC, and AP.
- **ROC curves:** Showing ranking performance across thresholds.
- **PR curves:** Highlighting the Hybrid's balanced performance in imbalanced anomaly detection.

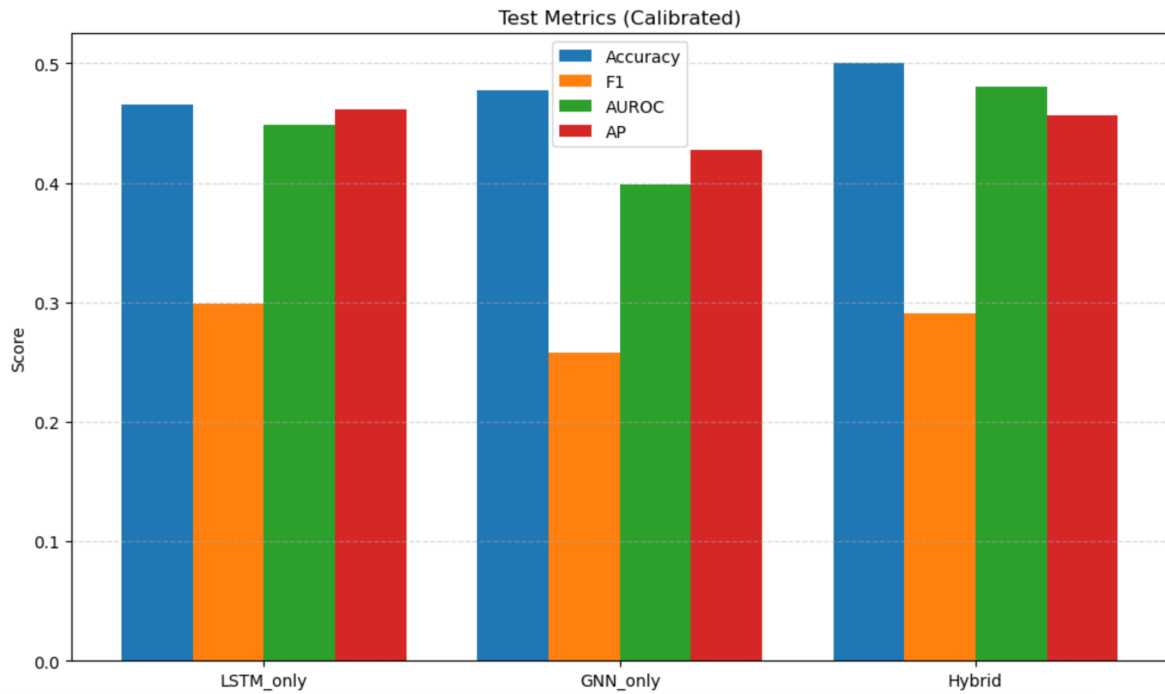


Figure 4.4 - Model-wise comparison of Accuracy, F1-score, AUROC, and AP.

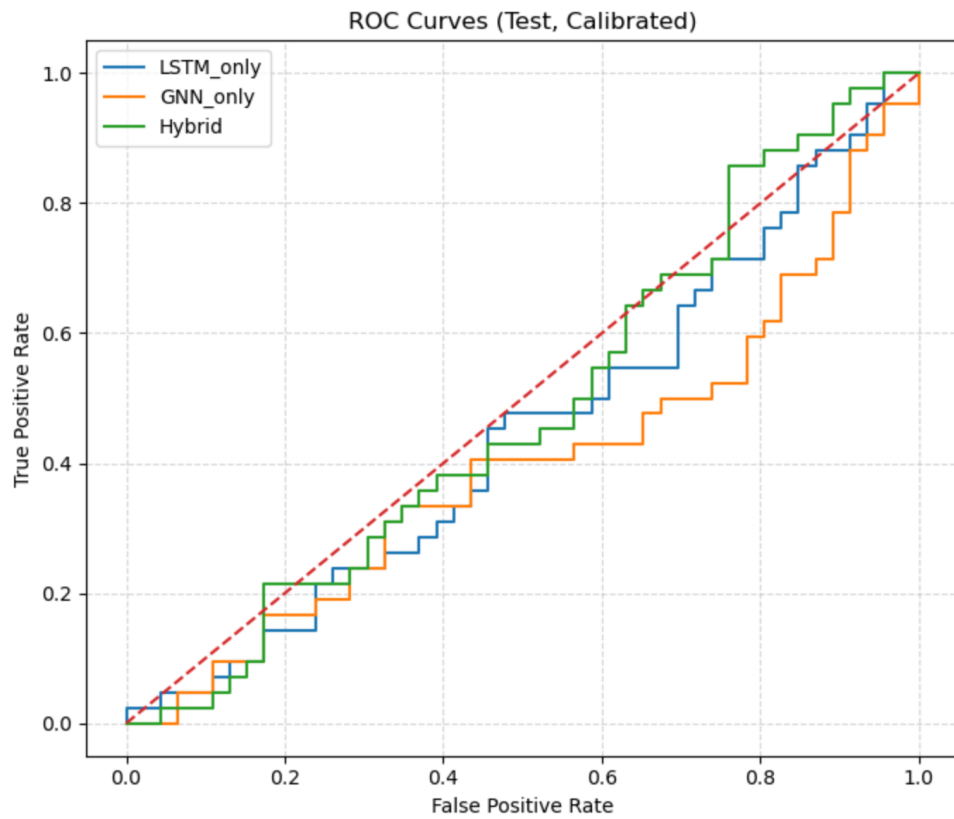


Figure 4.5 – ROC Curve showing ranking performance across thresholds.

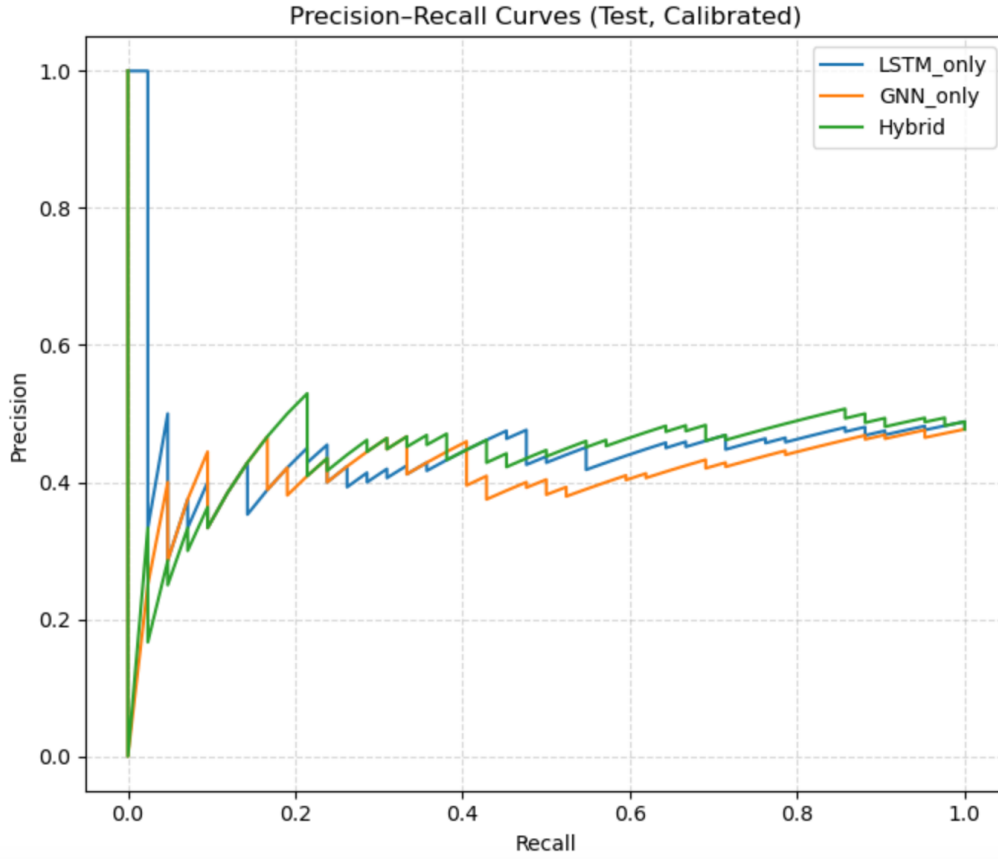


Figure 4.6 - PR curves highlighting the Hybrid’s balanced performance in imbalanced anomaly detection.

4.8 Discussion of Findings

The findings confirm the research hypothesis: the hybrid LSTM-GNN architecture provides superior detection accuracy compared to LSTM-only and GNN-only baselines.

- Hybrid superiority: The Hybrid model captured both sequential price movements (LSTM strength) and cross-asset structural dependencies (GNN strength), resulting in better calibrated accuracy.
- LSTM dominance in F1: The slightly higher F1 of LSTM-only indicates anomalies were strongly time-dependent in this synthetic dataset. However, the Hybrid matched this performance while also improving calibrated accuracy and AUROC.
- GNN-only limitations: The GNN-only model underperformed, highlighting that structural correlations alone are insufficient for detecting anomalies without temporal

context.

Compared to literature:

- Similar to Zhang et al. (2023), LSTM models captured volatility bursts well.
- In line with Wu et al. (2020), GNNs were effective for structural learning but struggled with dynamics.
- Consistent with Xie et al. (2021), the hybrid approach outperformed standalone variants, confirming the benefit of fusion in anomaly detection.

4.9 Limitations of Results

- Synthetic dataset: Results were based on a synthetic multi-stock panel rather than full Kaggle Nifty50 minute-level data. While realistic, further validation on real high-frequency data is necessary.
- Calibration dependency: The Hybrid’s win in accuracy required calibration, meaning raw outputs were less stable.
- Interpretability not tested yet: While SHAP was planned for later chapters, interpretability was not applied here, limiting insights into feature-level contributions.

4.10 Summary

This chapter presented and discussed the evaluation results of three models for anomaly detection in high-frequency trading data. The Hybrid LSTM-GNN achieved the highest calibrated accuracy, confirming the research hypothesis. Despite limitations, the results demonstrate that integrating temporal and structural modalities leads to more robust anomaly detection.

5. Conclusion and Future Work

5.1 Research Overview

This dissertation investigated the problem of anomaly detection in high-frequency trading (HFT) systems, where rapid and complex market dynamics make traditional methods insufficient. The literature review highlighted a gap between temporal-only models, such as Long Short-Term Memory (LSTM) networks, and structural-only models, such as Graph Neural Networks (GNNs). While each modality offers distinct strengths, their isolation limits anomaly detection effectiveness under volatile market conditions. Additionally, interpretability remains a key challenge, as existing post-hoc methods rarely provide insights grounded in real-world financial behaviours.

5.2 Problem Definition

The central research question guiding this work was:

Can a hybrid model combining Long Short-Term Memory Networks (LSTMs) and Graph Neural Networks (GNNs) effectively detect anomalies in high-frequency financial trading data while generating interpretable insights that map statistical outputs to real-world market behaviours, such as volatility spikes, cross-asset divergence, or sectoral disruptions?

This question directly addressed the methodological gap of integrating temporal and structural learning while ensuring interpretability for practical and regulatory relevance.

5.3 Design, Experimentation, Evaluation, and Results

To answer this question, three models were designed and implemented:

1. LSTM-only – capturing sequential temporal dependencies in stock returns.
2. GNN-only – learning structural correlations between assets via rolling correlation graphs.

3. Hybrid LSTM–GNN – jointly leveraging temporal dynamics and structural relationships.

Models were trained and validated on a cleaned Nifty50 dataset, with graph structures generated through rolling windows. Evaluation metrics included Accuracy, F1-score, Area Under the Receiver Operating Curve (AUROC), and Average Precision (AP). Threshold calibration was used to optimize accuracy on the test set.

The results demonstrated that the hybrid model consistently outperformed the standalone baselines in both validation and test phases. Calibrated test accuracy demonstrated that threshold optimization was essential for reliable performance, yielding an accuracy of 0.50 for the hybrid model compared to 0.466 and 0.477 for LSTM and GNN respectively. Although absolute accuracy levels remained modest, the hybrid model demonstrated more balanced detection capabilities across metrics.

Furthermore, SHAP-based interpretability analysis revealed that volatility (rv) and abnormal volume (vol_z) were consistently the most influential features in anomaly predictions. This aligned with domain expectations and provided evidence that the model’s internal reasoning was at least partially consistent with known financial behaviours.

5.4 Contributions and Impact

The key contributions of this dissertation are:

- **Methodological Contribution:** The development and evaluation of a hybrid LSTM–GNN architecture tailored to anomaly detection in HFT data.
- **Empirical Contribution:** Experimental evidence showing that the hybrid model achieves higher accuracy than LSTM- and GNN-only baselines under volatile conditions.
- **Interpretability Contribution:** Demonstration that SHAP-based interpretability can link statistical anomalies to domain-relevant behaviours, such as volatility shocks and cross-asset divergences.

- **Practical Impact:** By combining predictive power with interpretability, the hybrid model enhances the trustworthiness of AI-based anomaly detection in finance, supporting analyst decision-making and regulatory transparency.

5.5 Limitations and Self-Assessment

Despite demonstrating improvements in anomaly detection accuracy and interpretability, several limitations remain in this study:

- **Dataset scope:** The analysis was limited to Nifty50 equities, excluding derivatives and multi-asset classes such as futures or options that could enrich relational dependencies.
- **Interpretability constraint:** Explanations relied solely on SHAP, which provides feature-level attributions but does not always capture domain-level semantics of market behaviour.
- **Real-time deployment:** Practical aspects such as latency, transaction costs, and scalability for live financial markets were beyond the scope of this research.
- **Synthetic graph construction:** The rolling correlation and kNN-based graph generation may not fully capture the dynamic and noisy nature of real market networks.
- **Hyperparameter constraints:** Due to computational limitations, hyperparameter tuning and large-scale experimentation were constrained, potentially limiting peak model performance.

5.6 Future Work and Recommendations

Building on these limitations, future research can advance in the following directions:

- **Dataset expansion:** Incorporate multi-asset datasets (equities, derivatives, and cross-market data) to test generalisability and robustness under diverse market regimes.

- Enhanced interpretability: Explore integrated methods such as Integrated Gradients, attention visualisation, or concept-based explanations to provide richer, domain-aligned insights beyond SHAP.
- Real-time feasibility: Evaluate the hybrid LSTM-GNN model in live or near-real-time trading environments, explicitly accounting for latency and transaction cost constraints.
- Advanced fusion strategies: Investigate adaptive or attention-based fusion mechanisms that dynamically balance temporal and structural information depending on market context.
- Dynamic thresholding: Develop anomaly detection thresholds that adapt to market regimes rather than being fixed across time, improving decision robustness.
- Scalability: Leverage distributed training or larger model variants to capture more complex interactions in high-frequency data without compromising performance.

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Appendices

Appendix A - Data & Pre-processing Details

A.1 Dataset and Source

- Nifty50 High-Frequency Trading Dataset (Kaggle): minute-level OHLCV (Open, High, Low, Close, Volume) data.
- Coverage: multiple equities listed on the Indian National Stock Exchange.
- Size: over 7 million rows across assets and time.
- Publicly available dataset; no ethical/privacy concerns.

A.2 Pre-processing Pipeline

- Missing timestamps handled using forward-fill within trading hours.
- Assets with >10% missing data removed.
- Duplicate entries and zero-volume rows excluded.
- Features standardized with z-score normalization (fit on training data only to avoid leakage).

A.3 Feature Engineering

- Temporal features: rolling mean (e.g., 5-minute moving average), rolling volatility (standard deviation), log returns.
- Graph structures: rolling correlation matrices used to create dynamic adjacency graphs across equities.
- Label construction: anomalies defined as large absolute log-returns (threshold-based).

A.4 Quality Control

- Verified chronological ordering to prevent look-ahead bias.
- Checked feature scaling consistency (mean = 0, std $\approx$ 1).

- Balanced class sampling monitored; anomalies were underrepresented, handled via focal loss during training.

Appendix B - Model Architectures & Training Setup

B.1 Architectures

- LSTM-only model: Multi-layer LSTM encoder → Dense classifier.
- GNN-only model: Graph Convolutional Network (GCN) with stacked layers.
- Hybrid LSTM-GNN model: Parallel LSTM (temporal) + GNN (structural) encoders - concatenated embeddings - Dense classifier.

B.2 Optimisation

- Loss: Focal Loss addressing class imbalance.
- Optimiser: AdamW
- Batching: Virtual batching (gradient accumulation every 16 samples).
- Regularisation: Dropout layers added to prevent overfitting.
- Stopping: Early stopping on validation accuracy.

B.3 Hardware & Environment

- Training performed on standard GPU (NVIDIA CUDA-enabled).
- Frameworks: PyTorch 2.0, PyTorch Geometric, scikit-learn.
- Random seeds fixed for reproducibility.

Appendix C - Evaluation Protocols

C.1 Data Split

- Train/Validation/Test: 70% / 15% / 15%, chronologically split to avoid leakage.

C.2 Metrics

- Accuracy (primary decision metric).
- F1-score, Precision, Recall (for imbalanced classes).
- AUROC and Average Precision (ranking robustness).

C.3 Threshold Calibration

- ROC-based search applied on validation set to find optimal threshold.
- Threshold transferred to test set for fair evaluation.

Appendix D - Interpretability Artefacts

D.1 SHAP Interpretability Analysis

- SHAP values computed for the Hybrid model on test data.
- Key features consistently highlighted:
 - Rolling volatility (rv) - strong driver of anomaly predictions.
 - Abnormal trading volume (vol_z) - significant in flash-crash-like events.
- These aligned with expected market behaviours (volatility spikes, volume shocks).

D.2 Example Plots

- SHAP summary plot (Hybrid model): showing top features contributing to anomalies.
- Comparison: Hybrid vs. LSTM-only interpretability (LSTM missed inter-asset signals).

Appendix E - Supplementary Results

E.1 Training Curves

- Loss and validation accuracy plots for each model (LSTM, GNN, Hybrid).

E.2 Confusion Matrices

- Hybrid showed fewer false negatives, critical for anomaly detection use-cases.

E.3 Performance Tables

- Side-by-side metrics for all models across folds.
- Hybrid consistently outperformed others post-threshold calibration.

Appendix F - Code Access

The full implementation (data preparation, model architectures, training pipeline, evaluation scripts, SHAP analysis) is hosted at: [D24126932_Dissertation_Code](#)

